

Exercise 1 (Test for significance of regression):

Consider the model from Exercise 2 on Sheet 3:

$$\begin{aligned} wage = & \beta_0 + \beta_1 \cdot age + \beta_2 \cdot HSGrad + \beta_3 \cdot SomeCollege + \beta_4 \cdot CollegeGrad \\ & + \beta_5 \cdot AdvancedDegree + \beta_6 \cdot health.Verygood + \varepsilon. \end{aligned}$$

with

$$\hat{\beta}'_{LS} = (52.61, 0.62, 11.01, 23.16, 37.97, 62.63, 9.13)$$

We want to test the following hypothesis:

$$H_0 : (\beta_1, \beta_2, \beta_3, \beta_4, \beta_5, \beta_6)' = \mathbf{0}$$

- (a) Specify the alternative hypothesis H_1 and describe both hypotheses in plain English.
- (b) Specify the model equation under H_0 .
- (c) Specify the restriction matrix $\mathbf{C}$.
- (d) The coefficient of determination for the full model is $R^2 = 0.2685$. Calculate the F -statistic.
- (e) What is the decision of the test at the significance level $\alpha = 0.05$? Use the appropriate quantile of the correct F -distribution from the table below. How many degrees of freedom does the distribution have? Interpret the result.

Table 1: Quantiles for some F -distributions

quantile	0.025	0.05	0.1	0.9	0.95	0.975
$df_1 = 1$	0.0010	0.0039	0.0158	2.7072	3.8446	5.0289
2	0.0253	0.0513	0.1054	2.3044	2.9987	3.6934
3	0.0719	0.1173	0.1948	2.0856	2.6079	3.1205
4	0.1211	0.1776	0.2659	1.9467	2.3749	2.7901
5	0.1662	0.2290	0.3220	1.8492	2.2171	2.5707
6	0.2061	0.2725	0.3672	1.7761	2.1016	2.4124
7	0.2413	0.3095	0.4046	1.7187	2.0126	2.2917

Exercise 2 (Composite test of a subvector):

For the model from Exercise 1 we now want to test the following hypothesis:

$$H_0 : \begin{pmatrix} \beta_1 \\ \beta_6 \end{pmatrix} = \mathbf{0}$$

- (a) Specify the alternative hypothesis H_1 and describe both hypotheses in plain English.

- (b) Specify the model equation under H_0 .
- (c) Specify the restriction matrix $\mathbf{C}$.
- (d) Calculate the F -statistic. Use $\hat{\boldsymbol{\varepsilon}}'\hat{\boldsymbol{\varepsilon}} = 3819720$ and $\hat{\boldsymbol{\varepsilon}}'_{H_0}\hat{\boldsymbol{\varepsilon}}_{H_0} = 3995721$.
- (e) What is the decision of the test at the significance level $\alpha = 0.05$? Use the appropriate quantile of the correct F -distribution from the table above. How many degrees of freedom does the distribution have? Interpret the result.

Exercise 3 (Test of significance):

For the model from Exercise 1 we now want to test the following hypothesis:

$$H_0 : \beta_1 = 0$$

- (a) How is the test used in this scenario called?
- (b) Calculate the test statistic. Hint: Use Exercise 2 from Sheet 3.
- (c) What is the decision of the test at the significance level $\alpha = 0.05$? Use the appropriate quantile of the distribution from the table below. Which distribution was used? How many degrees of freedom does the distribution have? Interpret the result.
- (d) Calculate the 95%-confidence interval for β_1 . Hint: Use Exercise 2 from Sheet 3.
- (e) Calculate the prediction interval for the wage of a 50 year old man with an advanced degree but less than good health. Use Exercise 2 from Sheet 3 and $\mathbf{x}'_0(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0 = 0.0035$.

Table 2: Quantiles for the distribution

quantile	0.025	0.05	0.1	0.475	0.9	0.95	0.975
	-1.9608	-1.6454	-1.2818	-0.0627	1.2818	1.6454	1.9608